HHCKLA Buddhist Leung Chik Wai College

6A error analysis (2023)

Student No.: 17 Name: _____ Marks: / 42

Class: 6A Date: _____ Marked date: _____

What have you learned in this chapter? _____

What can you do to improve your work next time? _____

1. If $\frac{a+5b}{7a+2b} = \frac{1}{b+3}$, then $a =$

A. $\frac{4-b}{5b^2+13b}$.

B. $\frac{4+b}{5b^2+13b}$.

C. $\frac{5b^2+13b}{4-b}$.

D. $\frac{5b^2+13b}{4+b}$.

(DSE-2023 PII Q01)

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2. Let $g(x) = 13 - 5x^2$. If α is a constant, find $g(1 - 3\alpha)$.

A. $8 - 45\alpha^2$

B. $8 + 45\alpha^2$

C. $8 - 30\alpha + 45\alpha^2$

D. $8 + 30\alpha - 45\alpha^2$

(DSE-2023 PII Q08)

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3. The marked price of a jacket is 60% above its cost. A profit of $104 is made by selling the jacket at a discount of 25% on its marked price. Find the cost of the jacket.

A. $416

B. $520

C. $728

D. $832

(DSE-2023 PII Q11)

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4. The length of a side of a solid cube is $60\,\text{cm}$. The volume of a solid right circular cylinder is equal to the volume of the cube while the curved surface area of the circular cylinder is equal to the total surface area of the cube. Find the base radius of the circular cylinder.

A. $20\,\text{cm}$

B. $30\,\text{cm}$

C. $76\,\text{cm}$

D. $172\,\text{cm}$

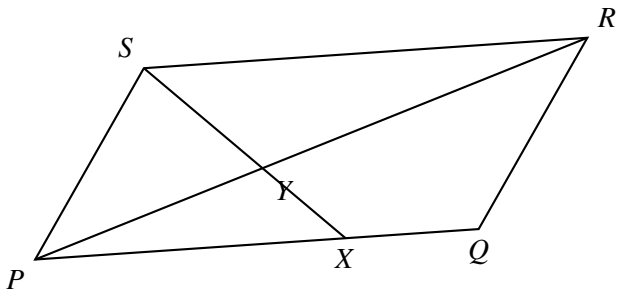
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5. In the figure, $PQRS$ is a parallelogram. Let X be a point lying on PQ. Denote the point of intersection of PR and SX by Y. If the area of $\triangle PXY$ and the area of the quadrilateral $QRYX$ are $32\,cm^2$ and $58\,cm^2$ respectively, then the area of $\triangle RSY$ is



A. $40\,cm^2$.

B. $50\,cm^2$.

C. $58\,cm^2$.

D. $72\,cm^2$.

(DSE-2023 PII Q17)

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6. It is given that $ABCD$ is a rhombus. Denote the point of intersection of AC and BD by E. Which of the following must be true?

I. $AE = BE$

II. $\frac{AE}{AC} = \frac{BE}{BD}$

III. $AE^2 + BE^2 = CD^2$

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

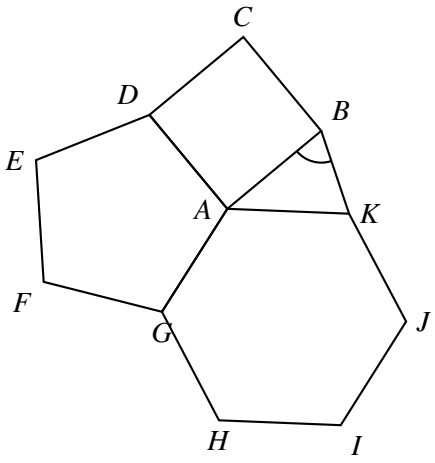
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7. The figure shows the square $ABCD$, the regular pentagon $ADEFG$ and the regular hexagon $AGHIJK$. Find $\angle ABK$.



A. 69°

B. 72°

C. 74°

D. 75°

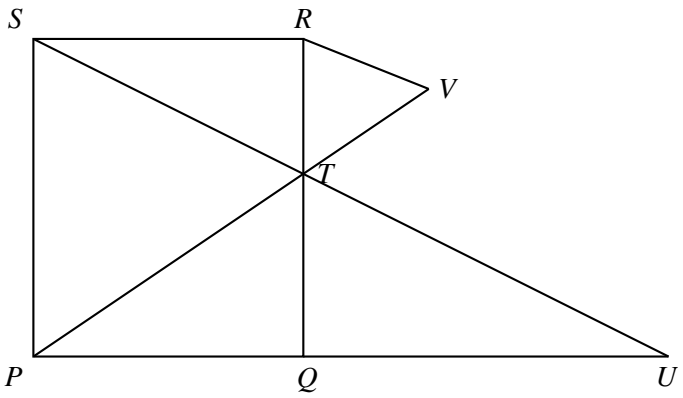
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8. In the figure, $PQRS$ is a rectangle. Let T be a point lying on QR such that $\angle PTS = 90°$. PQ produced and ST produced meet at the point U. PT is produced to the point V such that $RT = RV$. Which of the following must be true?



A. $RV \parallel ST$

B. $\angle PTQ = \angle RTS$

C. $\triangle PST \sim \triangle UTQ$

D. $\triangle PQT \cong \triangle TRS$

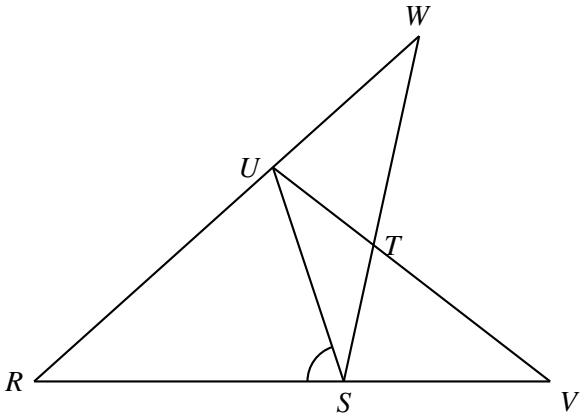
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9. The figure shows the cyclic quadrilateral $RSTU$, where $ST = TU$. RS produced and UT produced meet at the point V while RU produced and ST produced meet at the point W. If $\angle RWS = 32^\circ$ and $\angle RVU = 48^\circ$, then $\angle RSU =$



A. 65°.

B. 73°.

C. 80°.

D. 82°.

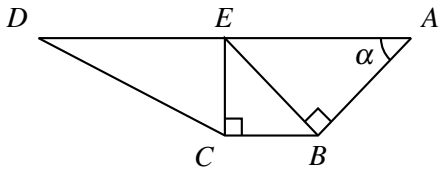
(DSE-2023 PII Q22)

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10. In the figure, $ABCD$ is a trapezium with $AD//BC$. Let E be the mid-point of AD. It is given that $\angle ABE = \angle BCE = 90°$. Find $\dfrac{CE}{DE}$.



A. $\dfrac{1}{2}$

B. 1

C. $\tan\alpha$

D. $\sin\alpha\cos\alpha$

(DSE-2023 PII Q23)

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11. The rectangular coordinates of the point P are $(\sqrt{2}, -\sqrt{2})$. If P is rotated anticlockwise about the origin through $90°$, then the polar coordinates of its image are

A. $(\sqrt{2}, 45°)$.

B. $(\sqrt{2}, 225°)$.

C. $(2, 45°)$.

D. $(2, 225°)$.

(DSE-2023 PII Q24)

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12. The equations of the straight lines ℓ and L are $9x+12y-37=0$ and $12x+16y+85=0$ respectively. ℓ cuts the x-axis at the point A while L cuts the y-axis at the point B. Let P be a moving point in the rectangular coordinate plane such that the perpendicular distance from P to ℓ is equal to the perpendicular distance from P to L. Denote the locus of P by Γ. Which of the following are true?

I. Γ is parallel to L.

II. Γ is perpendicular to AB.

III. Γ passes through the mid-point of AB.

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

(DSE-2023 PII Q26)

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13. The equations of the circles C_1 and C_2 are $x^2+y^2+7x-4y+15=0$ and $2x^2+2y^2-2x-16y-17=0$ respectively. Let G_1 and G_2 be the centres of C_1 and C_2 respectively. Denote the origin by O. Which of the following are true?

I. $\triangle OG_1G_2$ is an equilateral triangle.

II. The line segment OG_1 lies inside C_2.

III. C_1 and C_2 intersect at two distinct points.

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

(DSE-2023 PII Q27)

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14. It is given that $\log_8 y$ is a linear function of $\log_4 x$. The intercepts on the vertical axis and on the horizontal axis of the graph of the linear function are 5 and 3 respectively. Which of the following must be true?

A. $x^5y^2 = 8^{10}$

B. $x^6y^5 = 8^{20}$

C. $x^{10}y^3 = 8^{20}$

D. $x^9y^{10} = 8^{30}$

(DSE-2023 PII Q33)

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15. Let $f(x) = 3x^2 + 18mx + 22m^2$, where m is a real constant. Which of the following statements about the graph of $y = -f(3x)$ must be true?

I. The x-coordinate of the vertex of the graph is m.

II. The y-coordinate of the vertex of the graph is $5m^2$.

III. The equation of the axis of symmetry of the graph is $x + m = 0$.

A. I only

B. II only

C. I and III only

D. II and III only

(DSE-2023 PII Q35)

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16. Consider the following system of inequalities:

$$\begin{cases} x+3 \geq 0 \\ 2x+3y-12 \leq 0 \\ 5x-3y+12 \leq 0 \end{cases}$$

Let D be the region which represents the solution of the above system of inequalities. Find the range of values of β such that the greatest value of $\beta x+6y$ is 24, where (x, y) is a point lying in D.

A. $\beta \leq -10$

B. $\beta \geq -10$

C. $\beta \leq 4$

D. $\beta \geq 4$

(DSE-2023 PII Q37)

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17. The straight line $hx + ky = 6$ and the circle $x^2 + y^2 - 8x - 4y - 18 = 0$ intersect at the points M and N, where h and k are constants. If the coordinates of the mid-point of MN are $(1, 0)$, find k.

A. 4

B. 6

C. 9

D. 12

(DSE-2023 PII Q39)

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18. The base of the right pyramid $VABCD$ is the square $ABCD$. Let θ be the angle between $\triangle ABV$ and $\triangle BCV$. If $AB : AV = 5 : 4$, then $\cos\theta =$

A. $\frac{-25}{39}$.

B. $\frac{-17}{33}$.

C. $\frac{-9}{16}$.

D. 0.

(DSE-2023 PII Q40)

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19. The equations of the straight lines L_1 and L_2 are $3x-4y+k=0$ and $4x+3y-k=0$ respectively, where k is a positive constant. It is given that L_1 cuts the x-axis at the point P. Denote the point of intersection of L_1 and L_2 by Q. If R is a point lying on L_2 such that the in-centre of $\triangle PQR$ lies on the x-axis, then the x-coordinate of R is

A. $-7k$.

B. $-k$.

C. k.

D. $7k$.

(DSE-2023 PII Q41)

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20. There are 15 teachers in a group. If 5 teachers are selected from the group to form a committee consisting of 1 chairperson and 4 members, how many different committees can be formed?

A. 3003

B. 15015

C. 20475

D. 360360

(DSE-2023 PII Q42)

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21. It is given that n is an integer. Let u, v and w be the standard deviation, the median and the range of the group of numbers $\{1-9n, 3-9n, 4-9n, 5-9n, 7-9n\}$ respectively. Which of the following must be true?

I. $u = 2$

II. $v < 4$

III. $w > 6$

A. I only

B. II only

C. I and III only

D. II and III only

(DSE-2023 PII Q45)

I can finish _____ questions

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